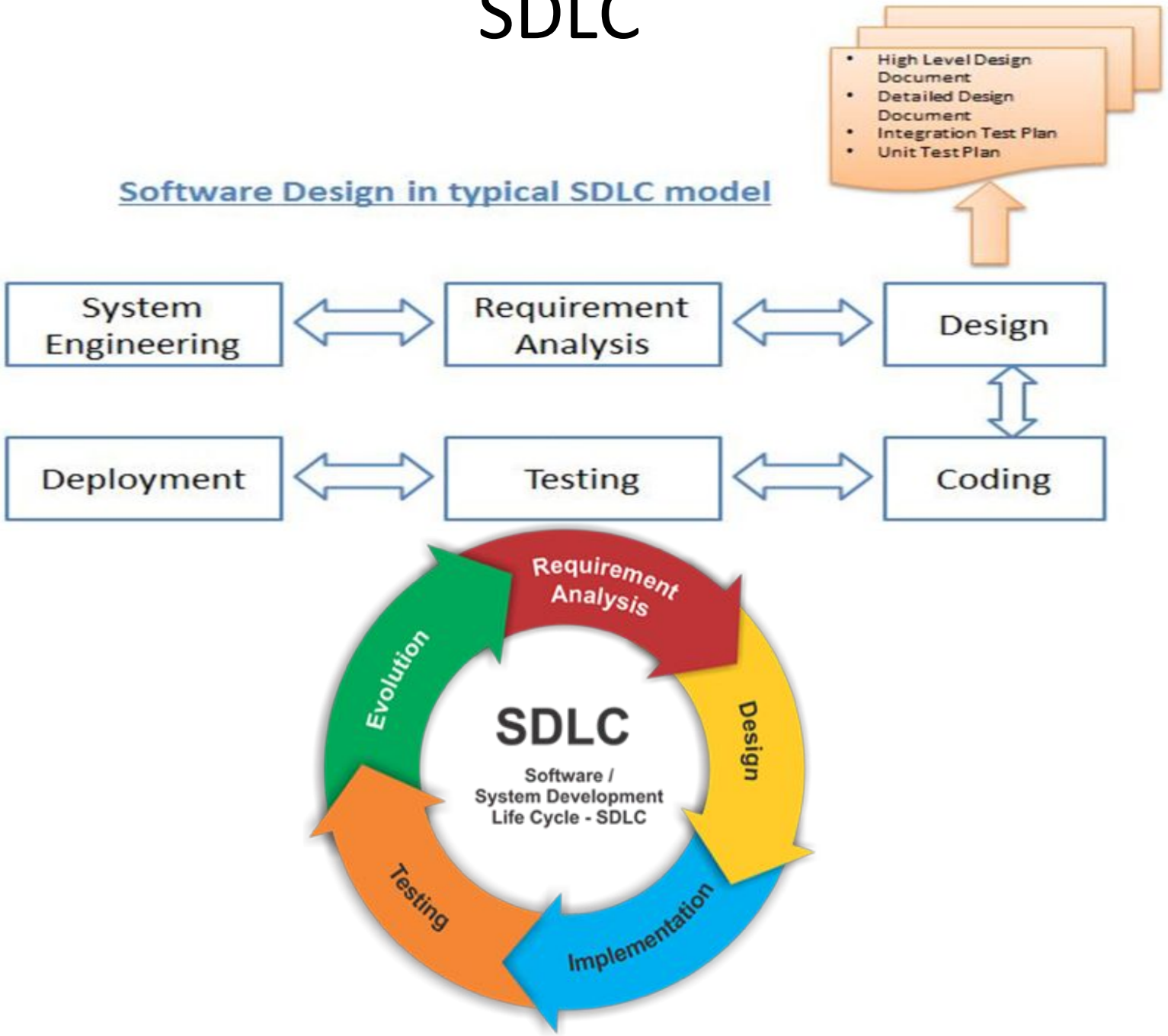
Software Design Process

SDLC


High Level Design Document
Detailed Design Document
Integration Test Plan
Unit Test Plan
Software Design in typical SDLC model
System Engineering
Requirement Analysis
Design
Deployment
Testing
Coding
Requirement Analysis
Design
Implementation
Testing
Evolution
SDLC
Software / System Development Life Cycle - SDLC

Design Process

- **Software Design** is a process to transform user requirements into some suitable form, which helps the programmer in software coding and implementation.

- The design phase of software development deals with transforming the customer requirements as described in the SRS documents into a form implementable using a programming language.

- The software design process can be divided into three levels of phases of design:
  - 1. Interface Design
  - 2. Architectural Design
  - 3. Detailed Design

Interface Design

- Interface design is the specification of the interaction between a system and its environment(user devices).
- During interface design, the internal of the systems are completely ignored and the system is treated as a black box.
- Attention is focused on the dialogue between the target system and the users, devices, and other systems(agents) with which it interacts.

Interface Design

Interface design should include the following details:

- Precise description of events in the environment, or messages from agents to which the system must respond.
- Precise description of the events or messages that the system must produce.
- Specification on the data, and the formats of the data coming into and going out of the system.
- Specification of the ordering and timing relationships between incoming events or messages, and outgoing events or outputs.

Architectural Design

- Architectural design is the specification of the major components of a system, their responsibilities, properties, interfaces, and the relationships and interactions between them.

- In architectural design, the overall structure of the system is chosen, but the internal details of major components are ignored.

Architectural Design

Issues in architectural design includes:

- Gross decomposition of the systems into major components.
- Allocation of functional responsibilities to components.
- Component Interfaces
- Component scaling and performance properties, resource consumption properties, reliability properties, and so forth.
- Communication and interaction between components.
- The architectural design adds important details ignored during the interface design. Design of the internals of the major components is ignored until the last phase of the design.

Detailed Design

- Design is the specification of the internal elements of all major system components, their properties, relationships, processing, and often their algorithms and the data structures.

Detailed Design

The detailed design may include:

- Decomposition of major system components into program units.
- Allocation of functional responsibilities to units.
- User interfaces
- Unit states and state changes
- Data and control interaction between units
- Data packaging and implementation, including issues of scope and visibility of program elements
- Algorithms and data structures

Design

- The main aim of design engineering is to generate a model which shows firmness, delight and commodity.

- Software design is an iterative process through which requirements are translated into the blueprint for building the software.

Software quality guidelines

1. A design is generated using the recognizable architectural styles and compose a good design characteristic of components and it is implemented in evolutionary manner for testing.
2. A design of the software must be modular i.e the software must be logically partitioned into elements.
3. In design, the representation of data , architecture, interface and components should be distinct.
4. A design must carry appropriate data structure and recognizable data patterns.

Software Quality Guidelines

5. Design components must show the independent functional characteristic.
6. A design creates an interface that reduce the complexity of connections between the components.
7. A design must be derived using the repeatable method.
8. The notations should be use in design which can effectively communicates its meaning.

Software Quality Attributes

The attributes of design name as “FURPS” are as follows:

1. **Functionality:** It evaluates the feature set and capabilities of the program.
2. **Usability:** It is accessed by considering the factors such as human factor, consistency and documentation
3. **Reliability:** it is evaluated by measuring parameters like frequency and security of failure, output result accuracy, recovery from failure and program predictability.

Software Quality Attributes

4. **Performance:** it is measured by considering processing speed, response time, resource consumption, throughput and efficiency.

5. **Supportability:** It combines the ability to extend the program, adoptability, serviceability, testability, compatibility and configurability are the terms using which a system can be easily installed and found the problem easily.

Supportability also consists of more attributes such as modularity, reusability, robustness, security, portability, scalability.